

## Circuit Simulation Lab:20

### *Design of Counters*

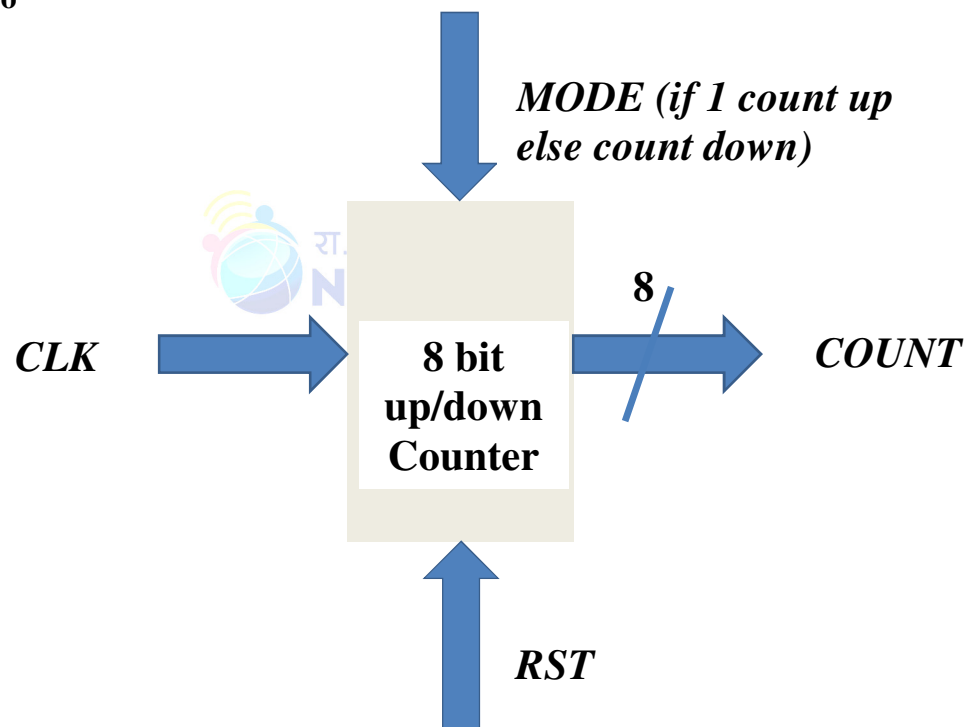
#### Introduction

The purpose of this experiment is to introduce the design of Synchronous sequential circuits, in this case we are taking an 8 bit up/down counter.

#### Software tools Requirement

**Modelsim (Siemens)**

**Xilinx Vivado**



**Synchronous 8 bit up/down Counter**

*Design the above behavior in Verilog HDL and write its testbench and capture the Waveforms*

*Behavior modeling*